



**TEMASEK JUNIOR COLLEGE  
PRELIMINARY EXAMINATION  
JC 2/ IP YEAR 6 2017**

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**H2 BIOLOGY**

Paper 2 Structured Questions

**9744/02**

**Tuesday 12 September 2017  
2 hours**

**READ THESE INSTRUCTIONS FIRST**

Write your name, Centre number, index number and class in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graph.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |      |
|--------------------|------|
| Q1                 | /10  |
| Q2                 | /10  |
| Q3                 | /12  |
| Q4                 | /11  |
| Q5                 | /12  |
| Q6                 | /12  |
| Q7                 | /11  |
| Q8                 | /12  |
| Q9                 | /10  |
| Total              | /100 |

This document consists of **24** printed pages and **2** blank pages.

**Paper 2 (Part I)**  
Answer **all** the questions in this section.

- 1 Fig. 1.1 shows an electron micrograph of a plant cell.

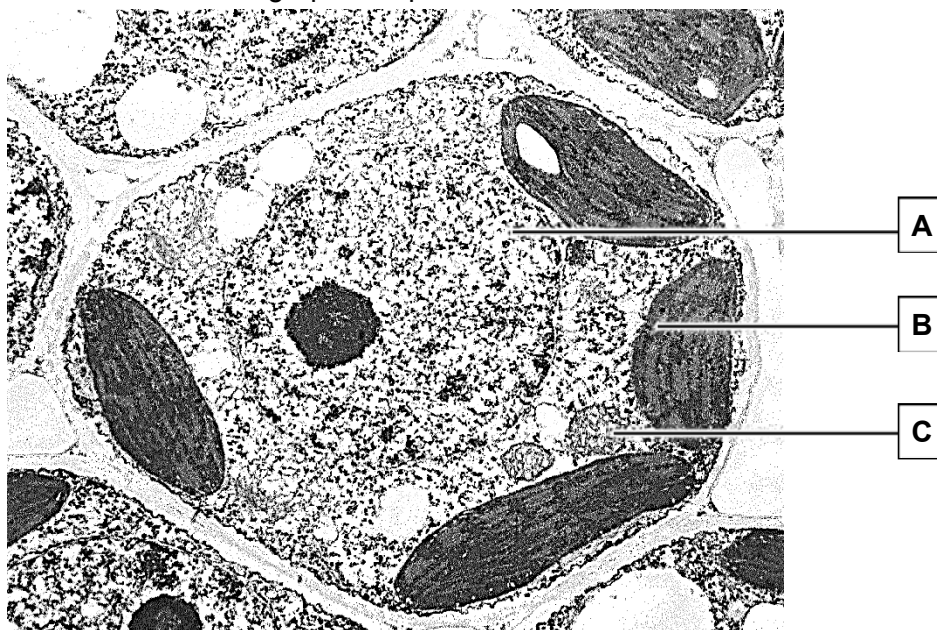


Fig. 1.1

- (a) Identify organelles **B** and **C**.

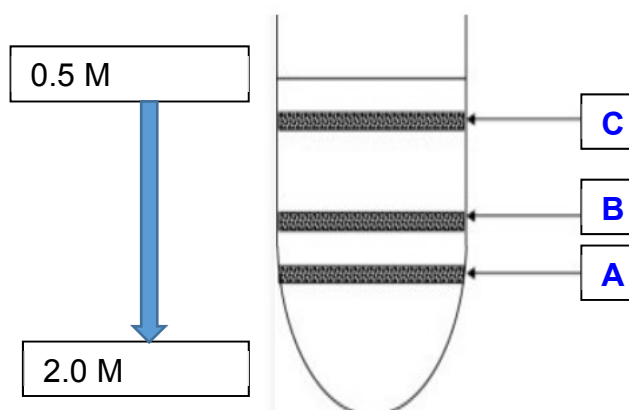
Organelle **B**: Chloroplast [1]

Organelle **C**: Mitochondrion [1]

- (b) Extracts from the homogenised plant cells in Fig. 1.1 were added to a sucrose density gradient and centrifuged at high speed to separate the various organelles.

- (i) Label the bands where organelles **A**, **B** and **C** can be found after centrifugation.

[3]



(ii) Explain your answer in (b)(i). [2]

**1. Density gradient**

**2. Organelles will separate according to their densities.**

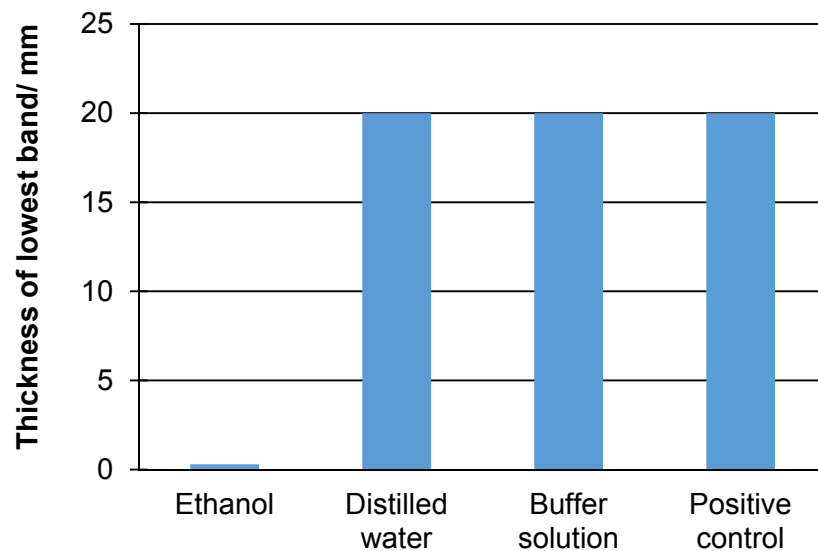
**3. Nucleus - heaviest**

**Chloroplast - medium size**

**Mitochondria – smallest size**

In a separate experiment, protoplasts (plant cells with cell wall removed) were first treated with three different reagents – ethanol, distilled water and buffer solution, for two hours. The treated cells were then subjected to the density gradient centrifugation.

Fig. 1.2 shows the thickness of the lowest band for each type of treated cell after density gradient centrifugation.



**Fig. 1.2**

(c) Explain the effects of the different reagents on the thickness of the lowest band. [3]

**0mm Ethanol – organic solvent – dissolves phospholipid bilayers thus no intact organelles (nucleus) can be obtained.**

**0mm Distilled water - Net movement of water molecules into nucleus, It has double membrane, therefore remained intact.**

**20mm Buffer solution –no net movement of water molecules, thus intact nucleus**

[Total: 10]

- 2 The membrane composition of bacterial species varies according to the environmental conditions to which the cells are exposed to, thus allowing them to thrive in a wide range of environment. Bacterial membranes consist of a large diversity of *amphiphilic* lipids, including sphingolipids (general structure is shown in Fig. 2.1).

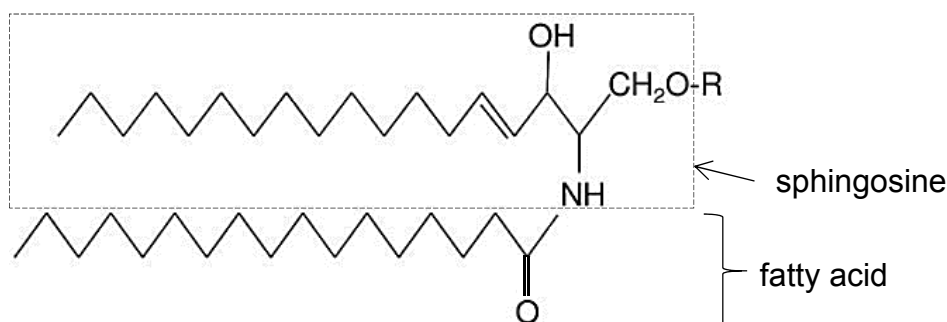


Fig. 2.1

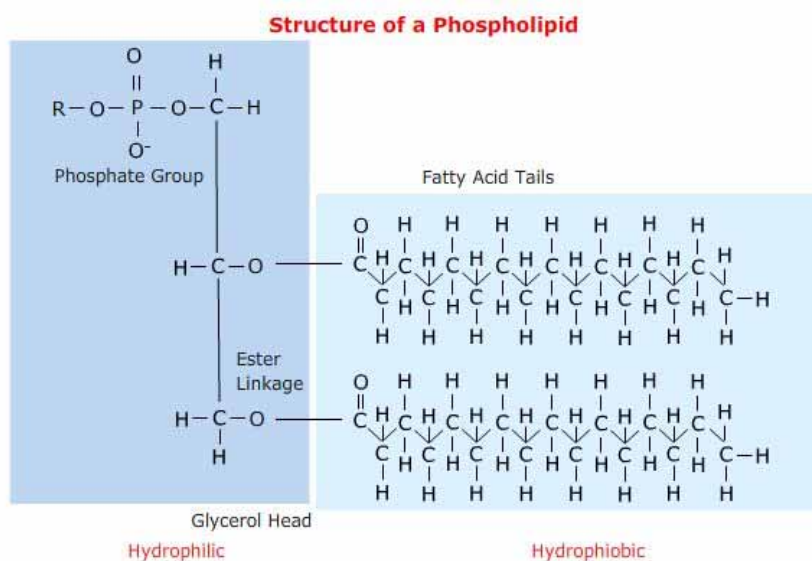
- (a) Define the term *amphiphilic*. [1]

Having both hydrophilic and hydrophobic parts

- (b) (i) State **two** differences between the structure of sphingolipids and phospholipids. [2]

| Sphingolipids                                               | Phospholipids                                          |
|-------------------------------------------------------------|--------------------------------------------------------|
| 1. <u>One fatty acid chain</u>                              | <u>Two fatty acid chains</u>                           |
| 2. <u>Presence of complex alcohol – sphingosine</u>         | <u>Presence of 3 carbon alcohol - glycerol</u>         |
| 3. <u>R group/ sugar attached to one end of sphingosine</u> | <u>Phosphate group attached to one end of glycerol</u> |

- (ii) Draw a labelled diagram of the molecular structure of a phospholipid. [3]

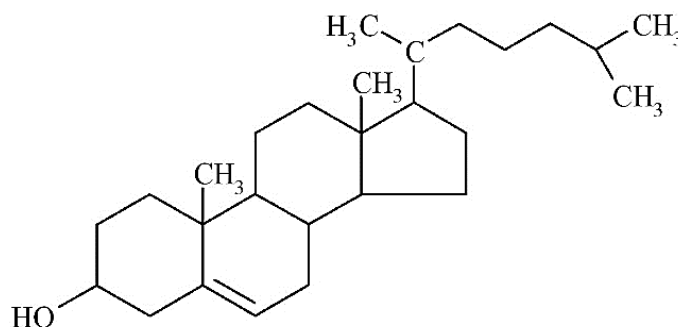


- **Labels: phosphate group; fatty acid tails; glycerol; ester bond; hydrophilic; hydrophobic**

(iii) Explain how the structure of phospholipid makes it a suitable component of cell membranes. [2]

1. Phospholipid molecules assemble into a bilayer.
2. The hydrophilic phosphate heads faces outwards and make contact with the aqueous environment on either side.
3. The hydrophobic tails face inwards and are sandwiched / buried between the hydrophilic heads.
4. Therefore, the hydrophobic region of the bilayer forms a boundary between the aqueous interior and exterior of the cell

Fig. 2.2 shows another component found in animal cell membranes.



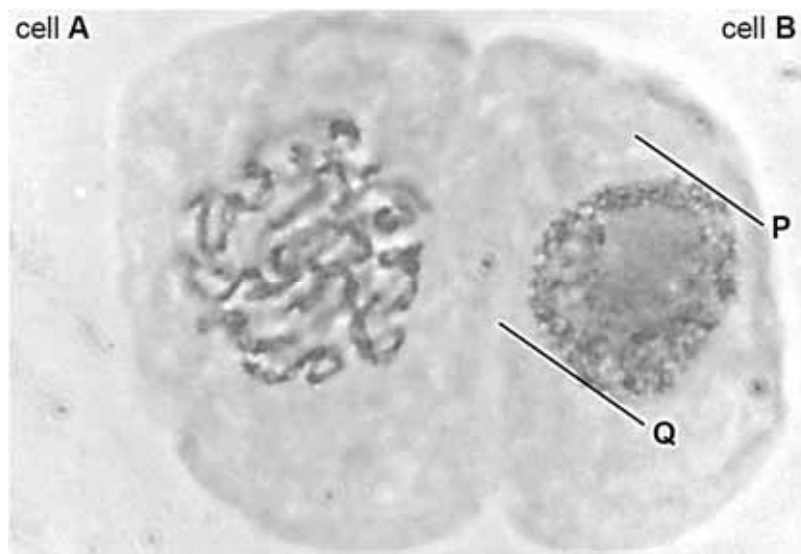
**Fig. 2.2**

(c) Explain how the molecule shown in Fig. 2.2 performs its function in cell membranes. [2]

1. At higher temperatures cholesterol reduce membrane fluidity.
2. At lower temperatures cholesterol helps prevent membranes from freezing by disrupting the close packing of phospholipids.

[Total: 10]

- 3 Fig. 3.1 shows a photomicrograph (Magnification x 5000) of two human cells, **A** and **B**, at different stages of the mitotic cell cycle.



**Fig. 3.1**

- (a) (i) Name the stage of mitosis that is occurring in cell **A**. [1]  
**Prophase (@early prophase)**

- 
- (ii) Describe the events that are occurring in cell **A**. [3]
1. **Chromosomes become visible due to condensation,**
  2. **Centrosome duplicates and migrate to opposite poles of the cell.**
  3. **The nucleolus disintegrates and nuclear envelope disintegrates.**
  4. **Spindle fibres extend from each pole towards the equator of the cell.**

- (b) Determine the actual length between lines **P** and **Q** in cell **B**. [2]

**Magnification = Image size / Actual size**

Allowance of  $\pm 1\text{mm}$  difference in measurement.  
 1 m for correct measurement AND working  
 1 m for correct answer WITH correct units.

Prostate cancer is one of the common cancers in males. It has been found that loss-of-function mutations to the DNA mismatch repair genes, such as *MSH2*, is associated with advanced prostate cancer. *MSH2* is a protein that dimerizes with another protein, *MSH6*, to form the DNA repair enzyme complex. *MSH2* has three domains – protein-binding domain, DNA-binding domain and ATPase domain (a region that removes phosphate group from ATP).

- (c) Explain how loss-of-function mutations in *MSH2* can lead to cancer development in the prostate. [3]
1. **need to occur to both alleles of the *MSH2* gene.**
  2. **The *MSH2* gene can be inactivated by:**
    - a) **A small chromosomal mutation or point mutation.**
    - b) **Its promoter region may become methylated.**

3. The mutations can lead to the inability of the MSH2 protein to:
- dimerize with MSH6
  - bind to the DNA / recognize incorrect/ damaged DNA sequences
  - hydrolyse ATP

Hormone therapy is typically used to treat prostate cancer but this treatment is only effective for a few years, after which resistance to the hormone will develop.

A study was carried out to determine the effectiveness of a chemotherapy drug, melphalan, in treating men with hormone-resistant prostate cancer. The side effect of melphalan is that formation of blood cells will be inhibited.

- (d) Suggest how the patient's own bone marrow can be used to counter the side effect of melphalan. [3]

- Collect and store hematopoietic stem cells from the patient's bone marrow
- Transfer/transplant the stem cells back into the patient
- The stem cells will proliferate and differentiate to the various types of blood cells in the body.

[Total: 12]

- 4 (a) Explain why mRNA is formed as a continuous strand during transcription while one of the DNA strands is formed discontinuously during replication. [3]

- DNA and RNA polymerases synthesize the new strands in the 5'→3' direction.
- template for DNA replication is double-stranded and anti-parallel, while template for mRNA synthesis is single-stranded.
- the direction of unwinding of the DNA template occurs opposite to the direction of synthesis for the lagging strand.

Several types of rRNA and tRNA are transcribed as a single strand precursor RNA. Each rRNA (16S, 23S, 5S) and tRNA molecule is cleaved following transcription in a process known as RNA trimming (Fig. 4.1) to form mature rRNA and tRNA molecules.

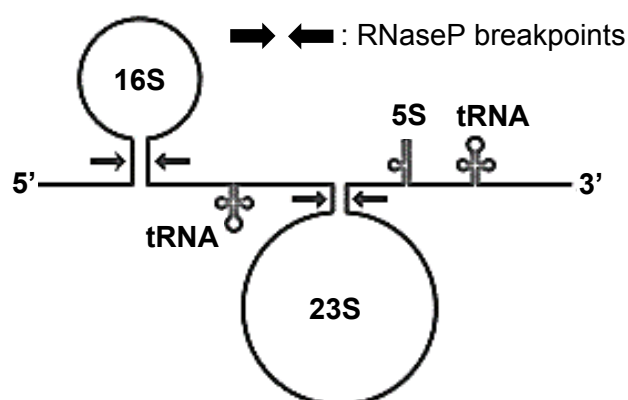


Fig. 4.1

- (b) State where rRNA genes are found. [1]  
**Nucleolus/ Mitochondria/ Chloroplasts**

- (c) Compare between the processes of RNA trimming and post-transcriptional modification for mRNA. [3]
1. **(Difference) Trimming – rRNA and tRNA are formed from a pre-RNA strand, whereas post-transcriptional modification for mRNA – only mature mRNA formed from pre-mRNA.**
  2. **(Difference) RNaseP is involved in trimming, whereas splicing involves spliceosome.**
  3. **(Similarity) both processes involve the removal of segments (e.g. intron for pre-mRNA) that are not required.**
- (d) Relate how the single-stranded structure of rRNA and tRNA facilitates their roles. [4]
1. **Single stranded structure –allow bases to fold back upon themselves, held in shape by hydrogen bonds between complementary base pairs**
  2. **rRNA - formation of the small ribosomal subunit, and the large ribosomal subunit.**
  3. **tRNA – formation of a structure that can fit into the E, P, A sites found on the large ribosomal subunit.**
  4. **Allows complementary base pairing of its anticodon with the codon of mRNA during translation to ensure that the correct sequencing of amino acids on the polypeptide chain.**

Total: 11]

## Paper 2 (Part II)

- 5 Operons in bacteria allow them to regulate their gene expression in response to changes in the environmental conditions.

In order to investigate the function of the regulatory and structural genes of *lac* operon, loss-of-function mutation was induced in the sequences of various genes. The different effects of the mutation on the expression of *lac* genes are shown in Table 5.1.

**Table 5.1**

| Region of DNA sequence in which gene mutation occurs | Allolactose absent     |                | Allolactose present    |                |
|------------------------------------------------------|------------------------|----------------|------------------------|----------------|
|                                                      | $\beta$ -galactosidase | transacetylase | $\beta$ -galactosidase | transacetylase |
| <b>A</b>                                             | +                      | +              | +                      | +              |
| <b>B</b>                                             | –                      | –              | –                      | –              |
| <b>C</b>                                             | –                      | –              | –                      | +              |
| <b>D</b>                                             | –                      | –              | +                      | –              |

(+) indicates the synthesis of functional enzyme

(–) indicates no synthesis of functional enzyme

- (a) (i) Identify regions **A** and **D**. [2]

**A:** *lacI*/ *lacI* promoter/ operator

**D:** *lacA*

- (ii) Outline the effect of the mutation of region **A** on the expression of *lac* genes. [2]

1. The *lac* repressor is not synthesized/ non-functional,

2. therefore it is unable to bind to the operator.

OR

1. Operator is mutated

2. *lac* repressor cannot bind to the operator.

3. Thus, regardless if the inducer allolactose is present or absent,

4. RNA polymerase is able to bind to the promoter to transcribe the genes of the *lac* operon.

Mammals respond to changes in the environmental conditions using different mechanisms. For instance, blood glucose concentration can be regulated by hormones such as insulin and glucagon.

Fig. 5.1 shows the modification of preproinsulin to form insulin in organelles **X** and **Y**.

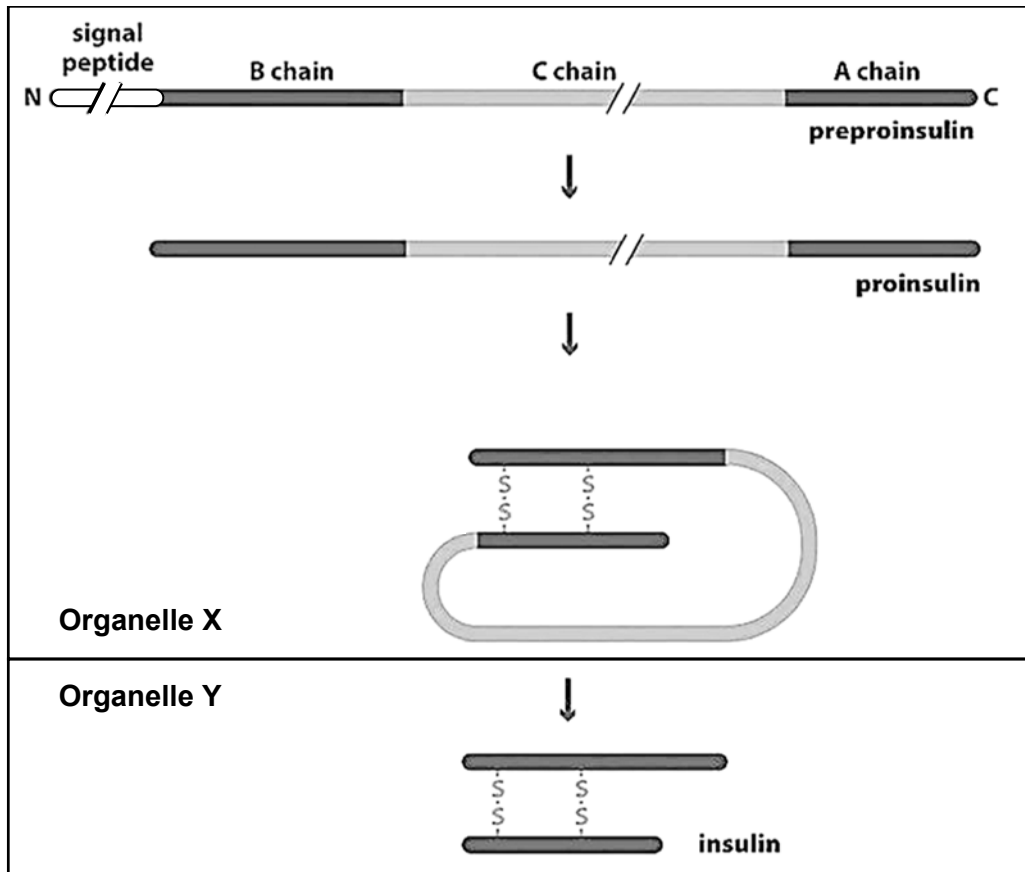


Fig. 5.1

(b) With reference to Fig. 5.1, outline what happens in organelles X and Y. [2]

**X: Rough endoplasmic reticulum**

1. The signal peptide of preproinsulin is cleaved in the rough endoplasmic reticulum to form proinsulin.

2. Two disulfide bonds were formed between the A and B chains.

**Y: Golgi apparatus**

3. The C-chain is cleaved/ hydrolyzed by (proteolytic) enzymes to form a functional insulin.

Fig. 5.2 shows the effect of glucose on a pancreatic cell.

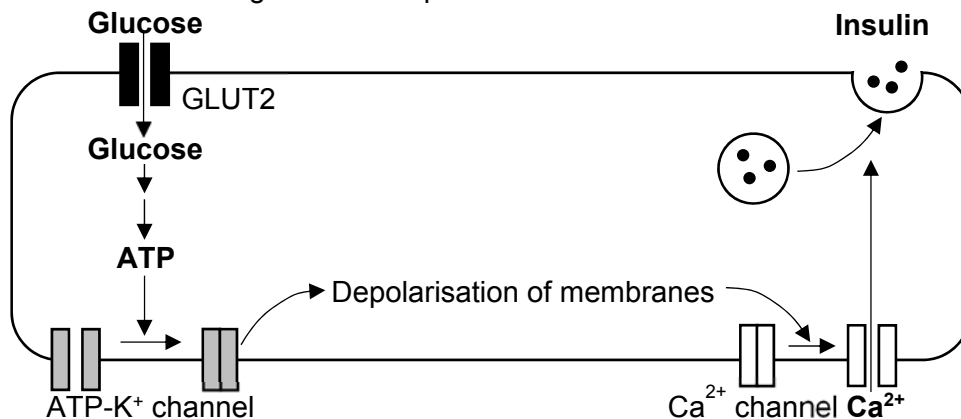


Fig. 5.2

- (c) With reference to Fig. 5.2, outline how the pancreatic cell responds to elevated blood glucose levels. [3]

1. Glucose enters the  $\beta$ -cells via facilitated diffusion through GLUT2.
2. Glucose is broken down into ATP.
3. The binding of ATP to ATP (sensitive)-K<sup>+</sup> channel closes the ATP-K<sup>+</sup> channel, causing the depolarization of the plasma membrane.
4. This triggers the opening of Ca<sup>2+</sup> channels, thus resulting in the influx of Ca<sup>2+</sup> into the cell.
5. triggers the fusion of insulin-containing secretory vesicles with the plasma membrane
6. to release insulin into the bloodstream.

Mammalian hormones can be synthesized artificially using bacterial cells.

- (d) Suggest **one** problem associated with expressing mammalian genes in bacterial cells. [1]

**ANY ONE:**

1. Introns are present as bacterial cells cannot carry out RNA splicing.
2. Eukaryotic promoter sequences / control elements may not be recognized by the bacteria, gene not expressed.

- (e) Compare the advantages of a mammalian response to changes in blood glucose concentration with that of a bacterial response to changes in supply of lactose. [2]

**Similarities**

1. Both allow the organism to utilise the increase in the supply of carbohydrates (glucose/ lactose).

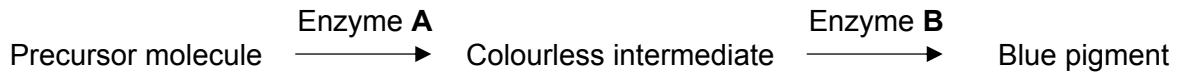
**Differences**

|                                      | Mammalian response                          | Bacterial response                           |
|--------------------------------------|---------------------------------------------|----------------------------------------------|
| 1. Rate of response                  | Respond <u>faster</u>                       | Respond <u>slower</u>                        |
| 2. Synthesis of proteins             | Hormones are <u>synthesized and stored.</u> | Enzymes are <u>synthesized when required</u> |
| 3. Regulation of carbohydrate supply | <u>Able</u> to regulate glucose supply      | <u>Unable</u> to regulate glucose supply     |

[Total: 12]

- 6** Harebell flowers can be found in various habitats in Scotland and Finland.

The colour of harebell depends on the anthocyanin pathway in which colourless intermediates are converted to a blue pigment. Two genes coding for two different enzymes, **A** and **B**, are crucial in this biochemical pathway. Each gene has two alleles, with the dominant alleles coding for functional enzymes while the recessive alleles coding for non-functional enzymes.



Two different white-petal homozygous lines of harebells were crossed and all the F<sub>1</sub> plants had blue flowers.

- (a) Using symbols **A** and **B**, draw a genetic diagram to explain the cross above and the result of selfing F<sub>1</sub>.

Key:

**A** represents the allele for the expression of petal colour

**a** represents the allele for preventing the expression of petal colour

**B** represents the allele for blue petals

**b** represents the allele for white petals

Parental phenotype: White x White

Parental genotypes : AAbb x aaBB

Gametes: (Ab) (aB)

F<sub>1</sub> genotype : AaBb  
F<sub>1</sub> phenotype : All blue flowers

Crossing F<sub>1</sub> : AaBb x AaBb

Gametes : (AB) (Ab) (aB) (ab) (AB) (Ab) (aB) (ab)

Punnett square

|      | (AB)                | (Ab)                 | (aB)                 | (ab)                 |
|------|---------------------|----------------------|----------------------|----------------------|
| (AB) | AABB<br>Blue flower | AABb<br>Blue flower  | AaBB<br>Blue flower  | AaBb<br>Blue flower  |
| (Ab) | AABb<br>Blue flower | AAbb<br>White flower | AaBb<br>Blue flower  | Aabb<br>White flower |
| (aB) | AaBB<br>Blue flower | AaBb<br>Blue flower  | aaBB<br>White flower | aaBb<br>White flower |
| (ab) | AaBb<br>Blue flower | Aabb<br>White flower | aaBb<br>White flower | aabb<br>White flower |

F<sub>2</sub> phenotypes : Blue flowers : White flowers  
F<sub>2</sub> phenotypic ratio: 9 : 7

[6]

The experimental results for the F<sub>1</sub> cross is shown in Table 6.1.

**Table 6.1**

|                  | Blue flowers | White flowers |
|------------------|--------------|---------------|
| Observed numbers | 12           | 9             |

A statistical test was carried out and a calculated value of 2.17 was obtained.

$\chi^2$  – Distribution Table

| Degrees of freedom | Probability, <i>p</i> |             |             |             |              |
|--------------------|-----------------------|-------------|-------------|-------------|--------------|
|                    | <b>0.10</b>           | <b>0.05</b> | <b>0.02</b> | <b>0.01</b> | <b>0.001</b> |
| 1                  | 2.71                  | 3.84        | 5.41        | 6.64        | 10.83        |
| 2                  | 4.61                  | 5.99        | 7.82        | 9.21        | 13.82        |
| 3                  | 6.25                  | 7.82        | 9.84        | 11.35       | 16.27        |
| 4                  | 7.78                  | 9.49        | 11.67       | 13.28       | 18.47        |

*t*-table

| Degrees of freedom       | Probability, <i>p</i> |             |              |             |              |
|--------------------------|-----------------------|-------------|--------------|-------------|--------------|
|                          | <b>0.1</b>            | <b>0.05</b> | <b>0.025</b> | <b>0.01</b> | <b>0.005</b> |
| <b>One-tailed t-test</b> |                       |             |              |             |              |
| <b>Two-tailed t-test</b> | <b>0.2</b>            | <b>0.10</b> | <b>0.050</b> | <b>0.02</b> | <b>0.010</b> |
| 1                        | 3.078                 | 6.314       | 12.706       | 31.821      | 63.657       |
| 2                        | 1.886                 | 2.920       | 4.303        | 6.965       | 9.925        |
| 3                        | 1.638                 | 2.353       | 3.182        | 4.541       | 5.841        |
| 4                        | 1.533                 | 2.132       | 2.776        | 3.747       | 4.604        |
| 5                        | 1.476                 | 2.015       | 2.571        | 3.365       | 4.032        |
| 6                        | 1.440                 | 1.943       | 2.447        | 3.143       | 3.707        |
| 7                        | 1.415                 | 1.895       | 2.365        | 2.998       | 3.499        |
| 8                        | 1.397                 | 1.860       | 2.306        | 2.896       | 3.355        |
| 9                        | 1.383                 | 1.833       | 2.262        | 2.821       | 3.250        |
| 10                       | 1.372                 | 1.812       | 2.228        | 2.764       | 3.169        |
| 11                       | 1.363                 | 1.796       | 2.201        | 2.718       | 3.106        |
| 12                       | 1.356                 | 1.782       | 2.179        | 2.681       | 3.055        |
| 13                       | 1.350                 | 1.771       | 2.160        | 2.650       | 3.012        |
| 14                       | 1.345                 | 1.761       | 2.145        | 2.624       | 2.977        |
| 15                       | 1.341                 | 1.753       | 2.131        | 2.602       | 2.947        |
| 16                       | 1.337                 | 1.746       | 2.120        | 2.583       | 2.921        |
| 17                       | 1.333                 | 1.740       | 2.110        | 2.567       | 2.898        |
| 18                       | 1.330                 | 1.734       | 2.101        | 2.552       | 2.878        |
| 19                       | 1.328                 | 1.729       | 2.093        | 2.539       | 2.861        |
| 20                       | 1.325                 | 1.725       | 2.086        | 2.528       | 2.845        |
| 21                       | 1.323                 | 1.721       | 2.08         | 2.518       | 2.831        |

(b) State a suitable conclusion for the experiment. [3]

1. For 1 degree of freedom, the calculated  $\chi^2$  value of 2.17 is less than 3.84,
2.  $p$ -value > 0.05.
3. Difference between the expected and observed numbers is statistically insignificant, and due to chance.
4. Experimental results followed the expected 9:7 ratio.

(c) Explain the genetic basis for the observed results. [3]

1. Complementary gene action
2. genotypes aaBB, aaBb or aabb have white petals.
3. Gene product of allele a masks phenotypic expression of the gene B.
4. Plants with the dominant allele A and two recessive alleles b will have white petal colour.
5. the recessive allele b code for white petals.
6. Plants with dominant allele A and dominant allele B will have blue petal colour.

[Total: 12]

- 7 Glucose and fructose are two common fruit sugars used in winemaking. Another sugar used in the fermentation industry is sucrose. The effects of the three sugars on fermentation by yeast were investigated and the results are shown in Fig. 7.1 and Fig. 7.2.

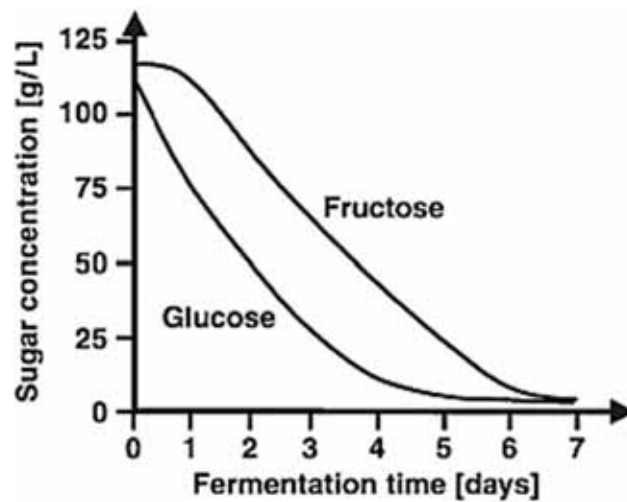


Fig. 7.1

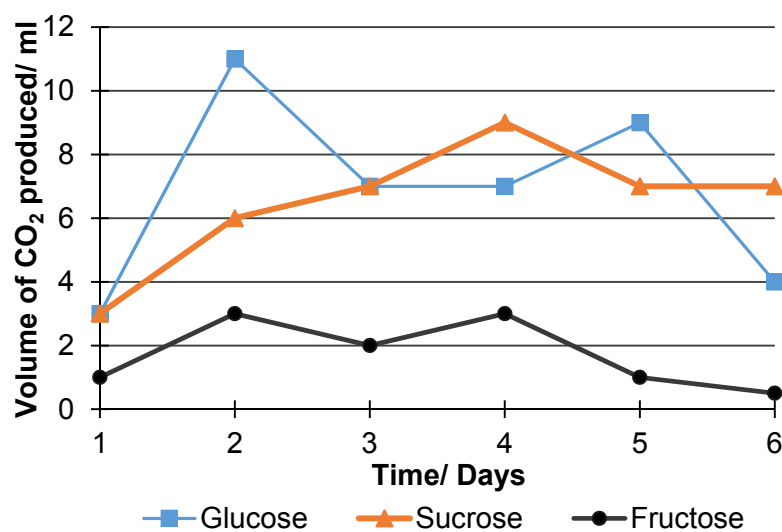


Fig. 7.2

- (a) Describe how ethanol is formed by yeast. [2]

1. Pyruvate is first decarboxylated to ethanal. The enzyme is pyruvate decarboxylase.

2. Ethanal is then reduced by NADH to form ethanol. NAD<sup>+</sup> is regenerated. The enzyme involved is alcohol dehydrogenase.

(b) With reference to Fig. 7.1 and Fig. 7.2, explain the order in which the sugars were utilized by yeast for fermentation. [4]

1. Glucose, then sucrose, followed by fructose.
2. Concentration of glucose decreased at a faster rate than concentration of fructose – at day 2, concentration of glucose dropped from 110 g/L to 50 g/L, as compared to concentration of fructose which dropped from 115 g/L to 90 g/L
3. CO<sub>2</sub> produced was also higher at 11 mL for glucose than for fructose (3 mL) and sucrose (6 mL)
4. These show that more glucose was used for fermentation, resulting in a higher level of CO<sub>2</sub> produced during the conversion of pyruvate to ethanal

Fig. 7.3 shows a respirometer.

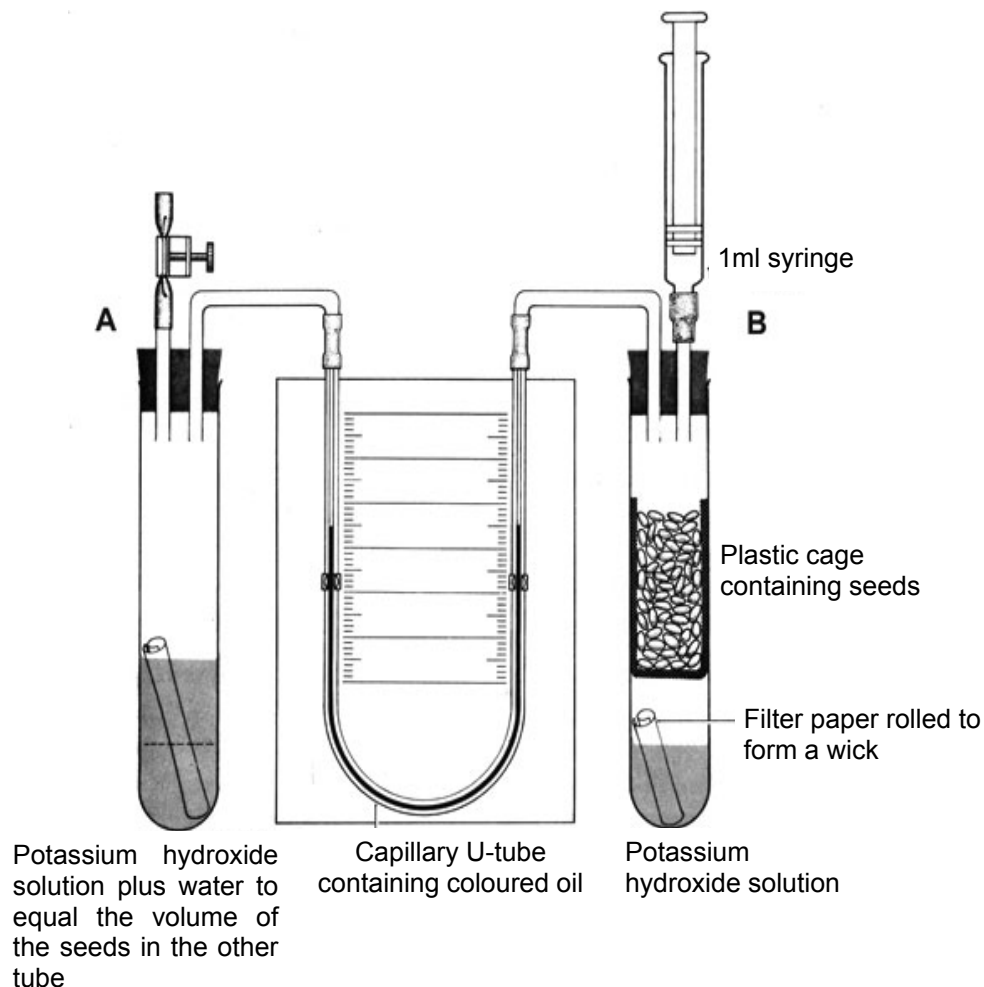


Fig. 7.3

(c) Briefly explain how you can determine the rate of respiration using the set-up shown in Fig. 7.3. [2]

1. Potassium hydroxide absorbs CO<sub>2</sub> produced
2. O<sub>2</sub> will be absorbed by the seeds
3. Thus, the level of the coloured oil will move up towards tube B
- 4.

(d) Suggest how the compensation point of a plant will be affected when it undergoes anaerobic respiration.

1. Compensation point is the amount of light intensity when rate of respiration corresponds with rate of respiration.
2. During anaerobic respiration, less CO<sub>2</sub> is produced as compared to aerobic respiration.
3. Compensation point will likely decrease

[Total: 11]

- 8 Human Immunodeficiency Virus (HIV) emerged as a mysterious new disease in the early 1980s.

Fig. 8.1 shows the structure of HIV.

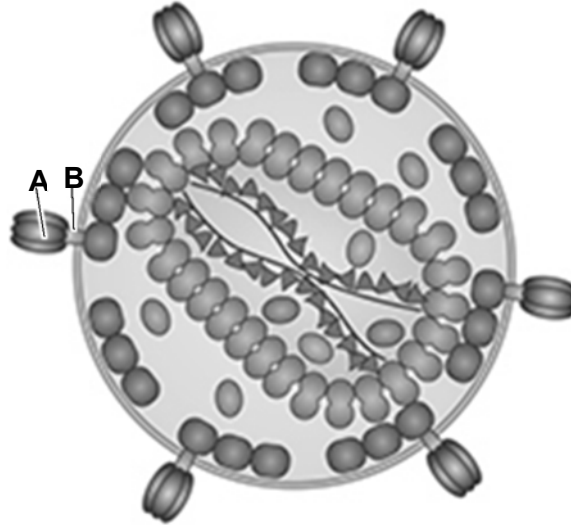


Fig. 8.1

- (a) Identify and describe the function of structures A and B. [3]

1. Structure A (gp120) binds to the CD4<sup>+</sup> receptor of T cells
2. Structure B (gp41) facilitates the fusion of viral envelope and plasma membrane of the CD4<sup>+</sup> T cells

In July 1990, a young woman in Florida, who had no known risk factors for HIV infection and no known contact with other HIV-positive persons, was tested HIV positive after undergoing an invasive dental procedure performed by a dentist who had Acquired Immunodeficiency Syndrome (AIDS).

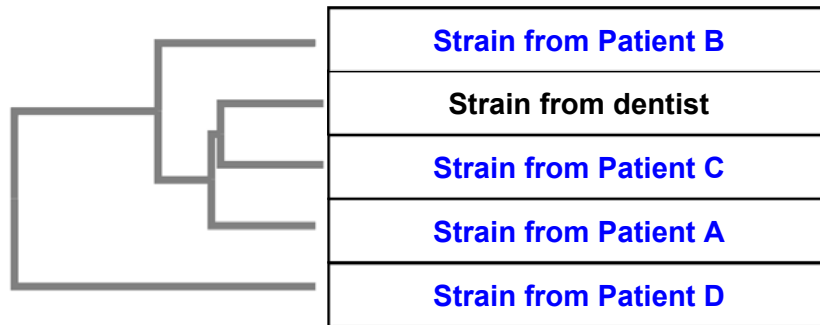
The U.S. Centre for Disease Control and Prevention (CDC) carried out an epidemiological investigation using DNA isolated from white blood cells of the dentist and patients A to D. The DNA was sequenced and compared with the strain isolated from the dentist.

Fig. 8.2 shows the multiple sequence alignment before a phylogenetic analysis could be carried out.

| Strain from | Sequence |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------------|----------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Dentist     | -        | - | - | - | C | - | T | A | - | T | T | G | - | C | T | G | G | C | G | C |
| Patient A   | -        | - | G | - | C | - | C | A | - | T | A | G | - | C | T | A | G | C | G | C |
| Patient B   | -        | - | G | - | C | A | C | C | - | T | - | G | - | C | T | A | G | C | G | C |
| Patient C   | -        | - | G | - | C | - | T | - | - | T | G | G | G | C | T | G | G | C | G | C |
| Patient D   | C        | A | G | A | C | - | T | A | C | T | - | G | - | C | T | A | G | - | G | - |

Fig. 8.2

- (b) Complete the figure below to show how closely related the different strains of HIV are.



[2]

In order to determine the origins of HIV, researchers conducted a similar phylogenetic analysis. Molecular epidemiology data showed that Simian Immunodeficiency Virus (SIV), which infects 36 species of primates found in sub-Saharan Africa, crossed over to infect human population known to eat wild animals. There are two types of HIV: HIV-1 and HIV-2.

Fig. 8.3 shows the phylogenetic tree of HIV / SIV.

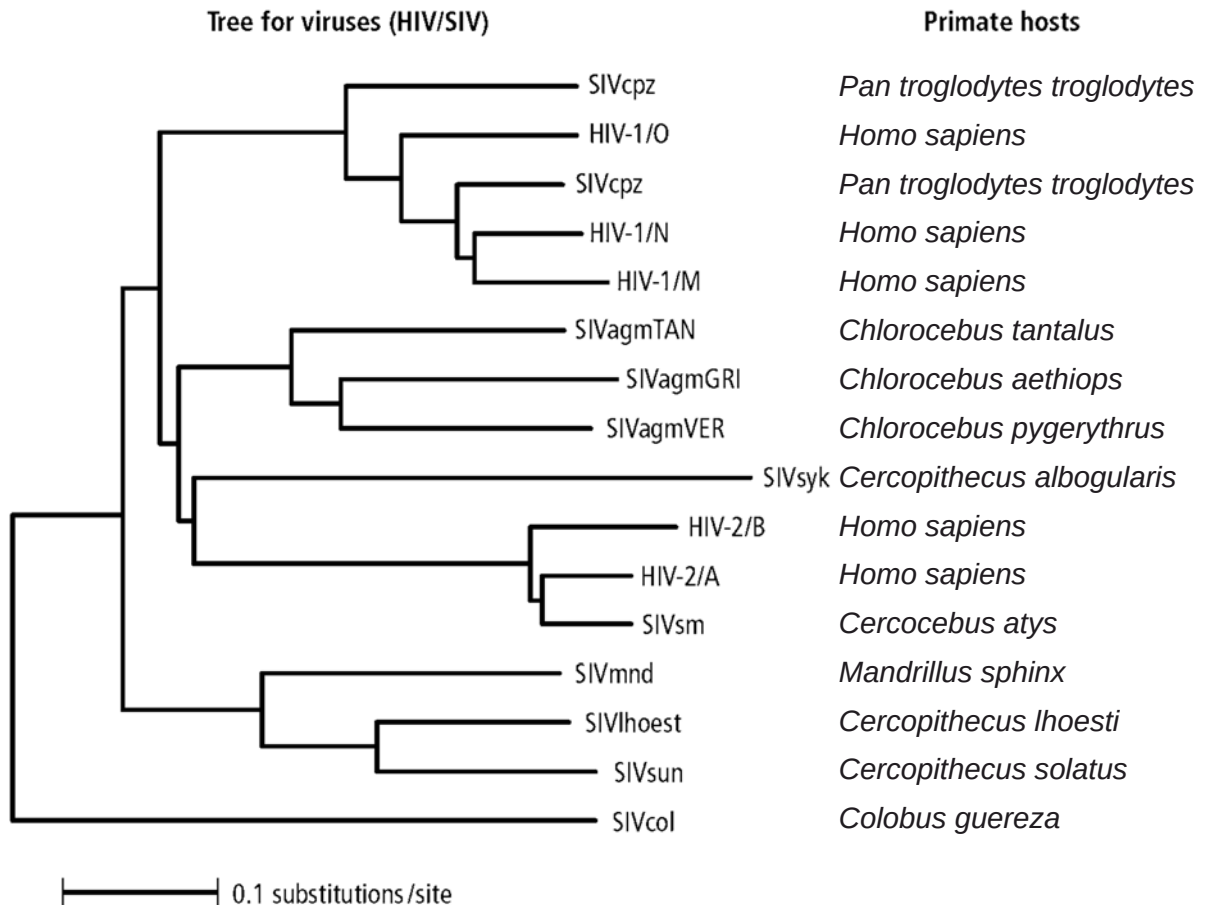


Fig. 8.3

(c) (i) Describe how HIV-1 and HIV-2 are evolutionarily related. [1]

1. They share/ arise from the common ancestor of SIV.

2. They arise independently/ are in different clades

(ii) Identify the closest viral strains and their primate hosts of HIV-1 and HIV-2. [2]

1. HIV-1 arises from SIVcpz from Pan troglodytes troglodytes [Chimpanzees]
2. HIV-2 arises from SIVsm from Cercocebus atys [Sooty mangabey]

SIV may have been around for more than a million years in these primates, but they do not cause disease in the primates. However, when SIV infected humans, it evolved to form HIV which causes AIDS in humans.

- (d) (i) Explain how HIV arose from SIV, allowing it to cause disease in humans. [3]
1. Genetic variation arises due to spontaneous random mutation.
  2. High mutation rate
  3. as a result of lack of proofreading ability of the reverse transcriptase
  4. Viruses with glycoproteins that are more complementary to the CD4 receptors of T helper cells
  5. are more infectious
  6. and they are more able to infect other hosts.
- (ii) Suggest the selective advantage for not causing death in the primate host [1].
1. Those viruses are able to replicate within the host and infect other hosts.

[Total: 12]

- 9 Tetanus is an infection caused by a Gram-positive bacterium, *Clostridium tetani*. The bacteria can exist in the dormant state as endospores. The endospores can be found everywhere in the environment, including soil, dust, and manure. They can enter the body through broken skin, usually through injuries from contaminated objects, and resume their active state.

- (a) (i) Describe the structure of the cell wall in *C. tetani*. [2]
1. thick peptidoglycan cell wall,
  2. a polymer of modified sugars (N-acetylglucosamine [NAG] and N-acetylmuramic acid [NAM]) cross-linked by short peptides.
- (ii) Suggest the advantage of forming of endospores. [1]
- Start to grow again when environment is suitable
- (b) Outline how the innate immune system responds to infection by *C. tetani*. [3]
1. Complement activation – formation of pores (MAC), enhanced phagocytic activity of macrophage
  2. Cytokine and chemokine secretion for recruitment of macrophages, neutrophils
  3. Inflammation – dilation of blood vessels

When *C. tetani* invades the body, a toxin is released into bloodstream. The toxin causes painful muscular contractions. It can also cause breathing problems, severe muscle spasms, seizures, and paralysis.

Tetanus vaccine was developed as a prophylactic measure against *C. tetani* toxins. The vaccine is produced by subjecting the bacterial toxins to chemical treatment to inactivate the toxins.

- (c) Explain how tetanus vaccine can prevent the clinical symptoms. [4]

1. Tetanus vaccine contains inactivated toxin that act as antigens.
2. The inactivated toxin is taken up by an antigen-presenting cell (APC) through phagocytosis.
3. The APC travels to a secondary lymphoid organ that contain mature naive T and B cells.
4. The antigens are processed by the APC and presented on the cell membrane of the APC by MHC class II glycoproteins.
5. Antigens on MHC class II are recognized by TCR on helper T cells.
6. These T lymphocytes become activated and proliferate.
7. Helper T cells interact and activate B cells,
8. which proliferate and differentiate to plasma cells to produce antibodies specific to the toxin.
9. Memory B and T cells are formed,
10. give long-term immunity against tetanus.

[Total: 10]